Literature-implied status: a subsymmetric direct-sum variant of Problem 8.1

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Status

This packet records a *literature-implied partial direct-sum variant*, not a full solution of the source problem. The source problem asks about arbitrary bases with the factorization property in a complemented sum. The later theorem used here proves a positive result for ℓ^p direct sums under stronger, subsymmetric structural hypotheses.

Source question

The source paper is Lechner–Motakis–Mueller–Schlumprecht, *Strategically reproducible bases and the factorization property* [1]. In Section 8, Problem 8.1 asks:

> 8. Final comments and open problems
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> Capon [4] showed that the bi-parameter Haar system in $L^p(L^q)$, $1 < p, q < \infty$, has the factorization property. With refined techniques, Capon's result was extended to $H^1(H^1)$ by [20] and then later in [13] to $H^p(H^1)$ and $H^1(H^p)$, $1 < p < \infty$. In Section 5, we gave a different proof of their results in, by writing $H^p(H^q)$ as a complemented sum of two spaces, solving the problem in each of the components separately, and then using the fact that strategically reproducibility is inherited by complemented sums. This begs the following question.
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> **Problem 8.1.** *If X and Y are Banach spaces with bases that have the factorization property, does the union of those two bases (in the right order) have the factorization property in the complemented sum of X and Y?*
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> As we remarked after Theorem 3.12 the uniform factorization property from

In words, if X and Y have bases with the factorization property, does the union of those bases, in the right order, again have the factorization property in the complemented sum of X and Y?

Supporting theorem

The supporting paper is Lechner, *Subsymmetric bases have the factorization property* [2]. Theorem 2.3 states:

Theorem 2.3 below can be viewed as a vector valued version of Theorem 2.2, but can also be regarded as an extension of the factorization result [17, Theorem 1.2] in the sense that [17, Theorem 1.2] is a corollary to Theorem 2.3 (see Corollary 5.1). This highlights the close connection between the factorization property of a basis and the primarity of the space. On the other hand, we would like to point out that although the James space is primary [8], the boundedly complete basis of the James space does not have the factorization property [20, Proposition 2.5].

Theorem 2.3. *For each $n \in \mathbb{N}$ let the dual pair $(X_n, Y_n, \langle \cdot, \cdot \rangle)$ of infinite dimensional Banach spaces X_n and Y_n and the sequences $(e_{n,j})_j$, $(f_{n,j})_j$ satisfy (B1)–(B5) with constant C_d (uniformly n). Assume that $(e_{n,j})_j$ is (C_u, C_s)-subsymmetric (uniformly in n) and non-ℓ^1-splicing and let $1 \le p \le \infty$. Then $(e_{n,j})_{n,j}$ has the factorization property in $\ell^p((X_n))$, i.e. whenever $T \colon \ell^p((X_n)) \to \ell^p((X_n))$ is a bounded operator with*

$$\inf_{n,j} |\langle T e_{n,j}, f_{n,j} \rangle| > 0,$$

there exist bounded operators $E, P \colon \ell^p((X_n)) \to \ell^p((X_n))$ such that $I_{\ell^p((X_n))} = PTE$.

For the proof of Theorem 2.3 we refer to Section 5.

The theorem applies to a sequence of Banach spaces X_n paired with Y_n and coordinate vectors $(e_{n,j})_j, (f_{n,j})_j$ satisfying the paper's hypotheses (B1)–(B5) uniformly. It further assumes that the coordinate bases $(e_{n,j})_j$ are uniformly (C_u, C_s)-subsymmetric and non-ℓ^1-splicing. Under these assumptions, the array $(e_{n,j})_{n,j}$ has the factorization property in $\ell^p((X_n))$ for every $1 \le p \le \infty$.

Idea of the proof

The source problem is hard because the factorization property need not behave formally under arbitrary decompositions of a space. Lechner's theorem supplies a controlled setting where the decomposition is an ℓ^p direct sum and the coordinate bases have enough symmetry to let a large diagonal operator be blocked while preserving a large diagonal on an array subbasis. The factorization operators are then built on the array basis itself, giving $I_{\ell^p((X_n))} = PTE$ for every operator T with large diagonal.

Precise packet theorem

Theorem 1. *Let $1 \le p \le \infty$. In the setting of Lechner's Theorem 2.3, namely for dual pairs $(X_n, Y_n, \langle \cdot, \cdot \rangle)$ satisfying* (B1)–(B5) *uniformly, and for uniformly subsymmetric, non-ℓ^1-splicing coordinate bases $(e_{n,j})_j$, the direct-sum array basis $(e_{n,j})_{n,j}$ has the factorization property in $\ell^p((X_n))$. Consequently, the ℓ^p direct-sum analogue of Problem 8.1 has a positive answer for this subsymmetric class of bases.*

Proof. This is exactly Theorem 2.3 of [2]. Given a bounded operator

$$T : \ell^p((X_n)) \longrightarrow \ell^p((X_n))$$

with large diagonal relative to the array basis, i.e.

$$\inf_{n,j} |\langle T e_{n,j}, f_{n,j} \rangle| > 0,$$

Theorem 2.3 gives bounded operators

$$E, P : \ell^p((X_n)) \longrightarrow \ell^p((X_n))$$

such that

$$I_{\ell^p((X_n))} = PTE.$$

This is precisely the factorization property for the array basis in the ℓ^p direct sum.

The relevance to Problem 8.1 is by direct identification: the array $(e_{n,j})_{n,j}$ is the ordered union of the coordinate bases in the direct sum model. The identification is only partial because Problem 8.1 assumes only that the individual bases have the factorization property and allows a general complemented-sum situation, while Theorem 2.3 assumes the stronger uniform subsymmetry and non-ℓ^1-splicing hypotheses and works in $\ell^p((X_n))$. $\square$

Verification notes and limitations

The packet uses a separate later paper and not a same-paper answer. The later paper does not explicitly state that it resolves Problem 8.1, so the result is classified as `literature_implied_answer`, not `literature_already_answered`.

The result should not be cited as a full answer to Problem 8.1. It leaves open the unrestricted complemented-sum question and the case of arbitrary bases with the factorization property. It also does not address the other open problems in Section 8 of [1].

References

[1] R. Lechner, P. Motakis, P. F. X. Mueller, and T. Schlumprecht. Strategically reproducible bases and the factorization property. *Israel Journal of Mathematics*, 238(1):13–60, 2020. arXiv:1809.09423.

[2] R. Lechner. Subsymmetric bases have the factorization property. arXiv:2011.09915, 2020.